Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building C++/Python systems, integration workflows, and hardware-adjacent software logic, with experience in debugging, validation, and supporting production-grade software-hardware interactions.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform integration services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting distributed APIs, automation workflows, and reliable release coordination across deployed environments.
- Maintained **20+ automated validation suites** with Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** and reducing regressions while strengthening defect isolation, reproducibility, and debugging across recurring build and integration cycles.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **backend services and validation workflows** for 500+ users, translating stakeholder requests into APIs, schema updates, and maintainable release pipelines while supporting production reliability and operational issue resolution across shared systems.
- Improved **system performance and reliability** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while strengthening debugging, traceability, and production support for deployed software.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Python services and control-style workflows** handling 10,000+ daily requests, integrating uploads, OCR processing, persistence logic, and automated checks to improve validation coverage, production reliability, and post-release support responsiveness.
- Debugged **runtime and workflow defects** across upload, OCR, persistence, and grading features, documenting fixes and reducing feedback turnaround by **60%** while improving issue isolation, troubleshooting speed, and maintainability across production systems.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **REST and WebSocket modules** over SQL/NoSQL persistence, supporting 2M+ synchronized records across integrations, dashboards, and operational workflows while strengthening high-volume reliability, service coordination, and software-component interoperability.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting speed, validation coverage, and engineering handoffs during debugging and iterative software updates.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **Python pipelines and evaluation workflows** across 100K+ trajectory logs, improving long-horizon planning by **18%** in simulation while strengthening testing discipline, algorithm validation, and reproducible engineering analysis.
- Architected a **spatial-temporal encoder** combining structured map geometry and agent context through multi-head attention, improving closed-loop planning stability by **15%** across repeatable validation, debugging, and simulation-analysis workflows.

Skills

Languages: **C++**, **Python**, Java, JavaScript, TypeScript, SQL, Bash
Systems: Linux/Unix, object-oriented design, data structures, debugging, validation, automation workflows, production support
Integration: REST APIs, WebSockets, CI/CD, SQL/NoSQL, Jenkins, Docker, AWS Glue/S3, Azure SQL, PostgreSQL
Hardware-Adjacent: control-style logic, software-hardware integration support, deterministic workflows, testing, troubleshooting

Projects

CPU Scheduling Simulator Live | *C++, React, JavaScript, OS Algorithms*
- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ** schedulers with C++ logic, debugging workflows, validation, timing analysis, and interactive visualization across simulated workloads, strengthening systems reasoning and correctness-focused testing practices.

HMIS Live | *MERN, Twilio, Sentry, TypeScript*
- Built a **full-stack operational platform** with appointment workflows, consent-based access, notifications, and monitoring, reducing administrative workload by **30%** while improving internal coordination, responsiveness, and day-to-day workflow clarity.

CaptionCraft Live | *Transformers, PyTorch, Flask, Computer Vision*
- Benchmarked **BLIP, GIT, ViT-GPT-2, and custom vision models** on 10K images and 100K captions, improving caption accuracy from **81%** to **96%** through repeatable evaluation pipelines and structured error analysis.